

① Basic Model

$$s > 0 \quad e > 0 \quad a > 0$$

$$x = s + e + a$$

Productivity $\rightarrow p(x) > 0$

Cost of effort $c(e) = \gamma e \quad \gamma > 0$

assumption $\rightarrow \limsup_{x \rightarrow \infty} \frac{p(x)}{x} < \gamma \quad \left\{ \begin{array}{l} \text{to ensure} \\ \text{monotonicity} \\ x^* \end{array} \right.$

$$u(e, s, a) = \frac{p(s+e+a)}{p(x)} - \gamma e$$

the agent chooses an effort level $e^*(s, a)$ to max u

$$e^*(s, a) = \max_{e > 0} u(e, s, a)$$

$$p^*(s, a) = p(s + e^*(s, a) + a)$$

↓ Solution

$$\max_{e > 0} u(e, s, a) = \max_{e > 0} \{ p(s+e+a) - \gamma e \}$$

$$e = x - s - a \text{ and } e > 0 \rightarrow x > s + a \text{ then}$$

$$= \max_{x > s+a} \{ p(x) - \gamma(x - s - a) \}$$

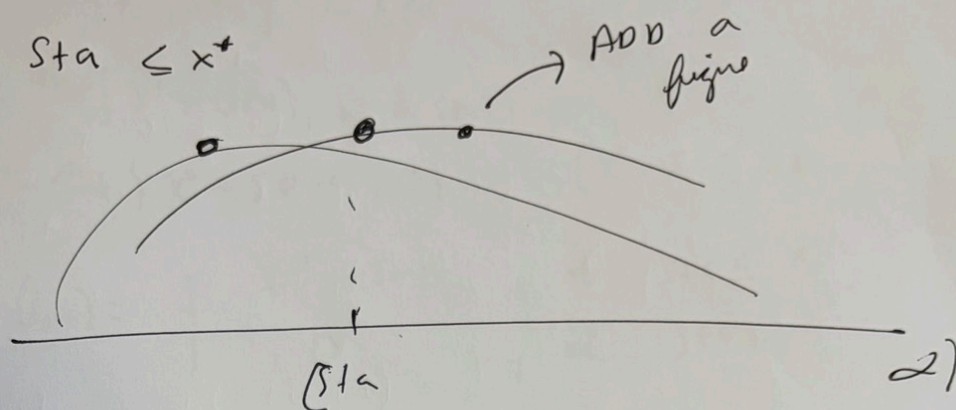
$$= \cancel{\gamma(s+a)} + \max_{x > s+a} \{ p(x) - \gamma x + \gamma s + \gamma a \}$$

$$= \gamma(s+a) + \max_{x > s+a} \{ p(x) - \gamma x \}$$

①

So we only need to focus on $p(x) - \gamma x$
on $[Sta, \infty]$

Case 1 $Sta \leq x^*$



Case 1

$Sta \leq x^*$

then the unconstrained the largest Maximum is
found at: $x = x^*$

$$e^* = x^* - s - a$$

$$p^* = p(x^*)$$

Case 2 $x^* < Sta$, If we are at the right of the
largest Maximum, the largest value
we should be $[Sta = x]$

If $Sta = x \rightarrow e = x - s - a = 0$ then
 $e^* = 0$, $p^* = p(Sta)$

$$e^*(s, a) = (x^* - s - a)^+$$

$$p^*(s, a) = \max \{ p(x^*), p(Sta) \}$$

$$e^*(s, a) = x^{opt} - (s + a)$$

$$= \max\{x^*, s + a\} - (s + a)$$

$$= \max\{x^* - (s + a), s + a - (s + a)\}$$

$$= \max\{x^* - s - a, 0\} = (x^* - s - a)^+$$

$$e^*(s, a) = \begin{cases} x^* - s - a & \text{if } s + a \leq x^* \\ 0 & \text{if } s + a > x^* \end{cases}$$

$$p^*(s, a) = p(s + e^*(s, a) + a)$$

$$= p(s + a + \max\{x^* - s - a, 0\})$$

$$= p(\max\{x^*, s + a\})$$

$\nearrow$ p is weakly increasing

$$p(\max\{x^*, s + a\}) = \max\{p(x^*), p(s + a)\}$$

$$p^*(s, a) = \begin{cases} p(x^*) & \text{if } s + a \leq x^* \\ p(s + a) & \text{if } s + a > x^* \end{cases}$$

$$e^*(s, a) = (x^* - s - a)^+,$$

$$p^*(s, a) = \max\{p(x^*), p(s + a)\}$$